

FOMC Meetings Analysis using NLP, Problem Statement & Background

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Business Problem

Financial institutions, investors, and analysts often face challenges in quickly extracting key insights from the lengthy and complex FOMC meeting minutes, which are critical for understanding the Federal Reserve's monetary policy direction. Delays or inefficiencies in analyzing these documents can lead to missed opportunities or suboptimal financial decisions, especially in fast-moving markets. With growing volumes of unstructured text data, there is a need for automated, accurate, and real-time analysis tools that can uncover sentiments, key topics, and trends from the minutes. This highlights a business problem where timely and actionable insights are crucial for staying competitive in investment, lending, and economic forecasting.

Corpus Viewer – FOMC Minutes (2022–2025)

We compiled a dataset of 26 FOMC meeting minutes, covering the period from January 2022 to March 2025, obtained from the Federal Reserve’s official website.

Using Orange’s Corpus Viewer, we ensured that:

- All documents were accurately loaded and labeled with their respective meeting dates
- The full text of each meeting was included to allow uniform and thorough NLP analysis
- The dataset is prepared for further analysis, including sentiment analysis, topic modeling, and trend identification

Corpus Viewer – Guardian News Articles (US Economy, 2022–2025)

With Orange’s Guardian widget, I built a corpus of over 19,000 U.S. economy-related news articles published between January 2022 and March 2025.

Each article includes:

- A headline
- The main body text
- Metadata like publication date, section (e.g., Business), and relevant tags

Word Cloud

FOMC Minutes (2022–2025)



- Dominant terms include **“market,” “inflation,” “rate,” “participant,” “price,” “bank,” and “reserve”**
- Frequent use of **economic indicators** like **“growth,” “credit,” “loan,” and “demand”** reflects the central role of inflation, employment, and monetary policy
- Phrases like **“expect,” “remain,” and “increas(e)”** point to forecasting and decision-making language

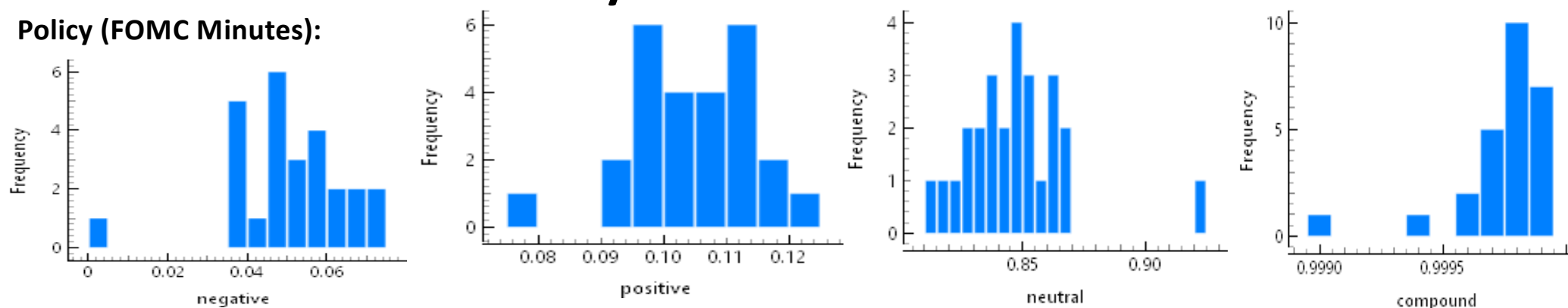
Guardian News Articles (US Economy, 2022–2025)



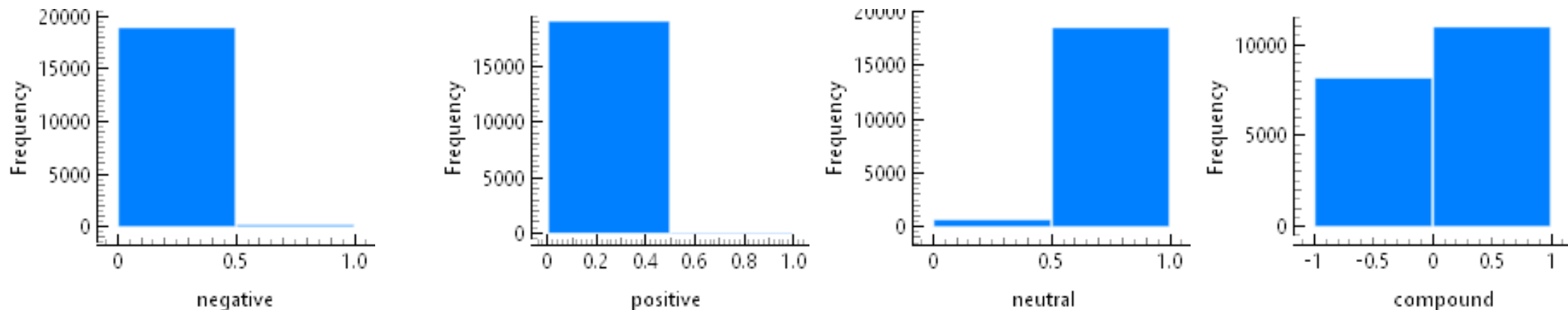
- Prominent terms include **“happen,” “say,” “Trump,” “UK,” “US,” “war,” “Ukraine,” and “inflation”**
- Topics extend beyond U.S. economic policy to include **global political events**, such as **Russia-Ukraine conflict, China relations, and climate change**
- Recurrent figures: **Trump, Biden, Sunak, Putin**, reflecting political discourse in economic reporting

Sentiment Trends in Policy vs. Media

Policy (FOMC Minutes):



Media (Guardian News):



Policy (FOMC Minutes):

- Sentiment scores are **narrowly distributed** and cluster around:
 - Slightly **positive** (avg ~0.1)
 - **Highly neutral**, suggesting objective, formal language
- Very low negative sentiment → Reflects restrained tone typical of central bank communications

Media (Guardian News):

- Much broader distribution in all sentiment categories
- Noticeable **volume of negative and polarizing (± 1) compound scores**
 - Indicates stronger language, emotional framing
- Media tends to use **more extreme tone**, both positive and negative, reflecting public discourse

Topic Modeling with LDA

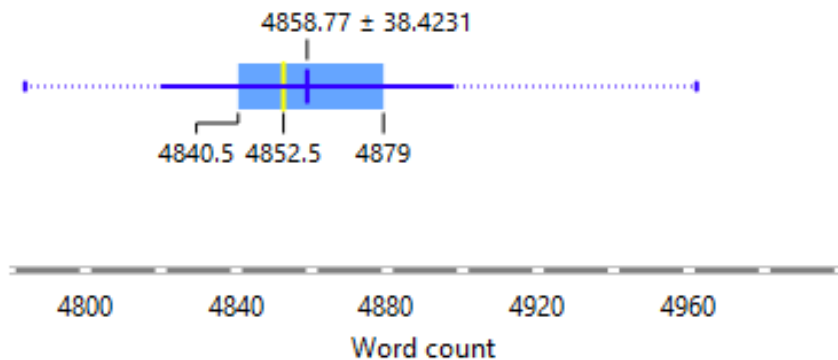
Policy (FOMC Minutes):

TOPIC	KEYWORDS (SHORTENED)	INTERPRETATION
1	particip, market, rate, inflat, price, risk	Inflation, risk assessment, interest rates
2	bank, reserv, committe, board, treasuri, secur	Institutional structure, Fed operations
3	foreign, market, author, currenc, transact	International markets & monetary policy
4	board, divis, expect, remain	Policy decision process and projections
5	particip, remain, feder	Participation & guidance language
6	credit, financi, price, polici	Financial conditions & policy direction

Media (Guardian News):

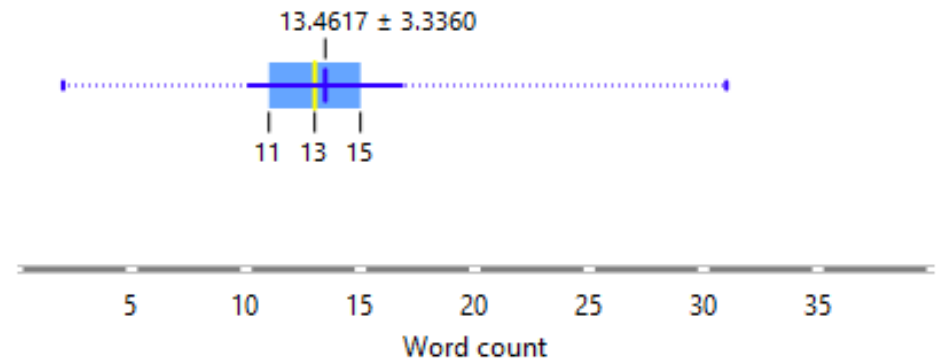
TOPIC	KEYWORDS (SHORTENED)	INTERPRETATION
1	happen, war, russia, ukraine, biden, strike	Russia–Ukraine conflict & global response
2	australia, world, nation, voic, morn	Global politics & international affairs
3	crisis, life, new, climat, nsw	Domestic crises, climate, and news alerts
4	uk, rate, rise, bank, inflat	Media framing of inflation & economic policy
5	sunak, tax, budget, tori, labour	UK political discourse & economic reform
6	trump, elect, biden, republican, senat	U.S. political narratives and elections

Document Complexity & Vocabulary



FOMC Minutes (Top Plot):

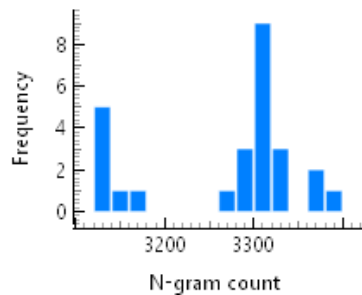
- Average length: approximately 4,859 words
- Narrow variance ($\sim \pm 38$ words)
- Suggests a formal, standardized format
- Demonstrates thorough discussion of policies, economic conditions, and decisions



Guardian Articles (Bottom Plot):

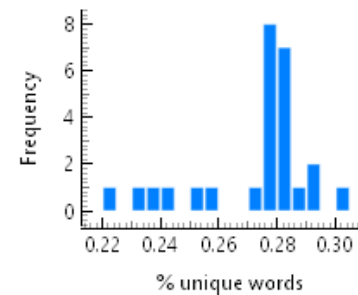
- Average excerpt length is around 13.5 words
- Significantly shorter and more variable
- Represents headlines or brief summaries, consistent with news-style snippets typically obtained through API extraction

Text Complexity & Lexical Diversity – Policy vs. Media



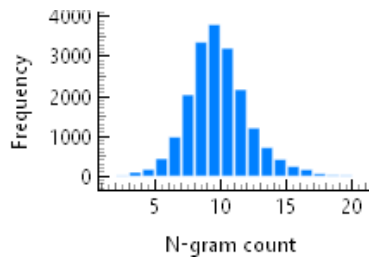
FOMC

- N-gram counts are notably high, clustering around 3,200–3,300
- Indicates frequent repetition of phrases, characteristic of formal institutional language (e.g., “Federal Reserve,” “interest rate”)



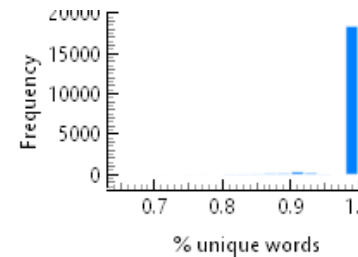
Guardian

- N-gram counts are significantly lower and more dispersed, averaging around 10
- Reflects the use of brief, diverse phrasing typical of headline-focused news content



FOMC

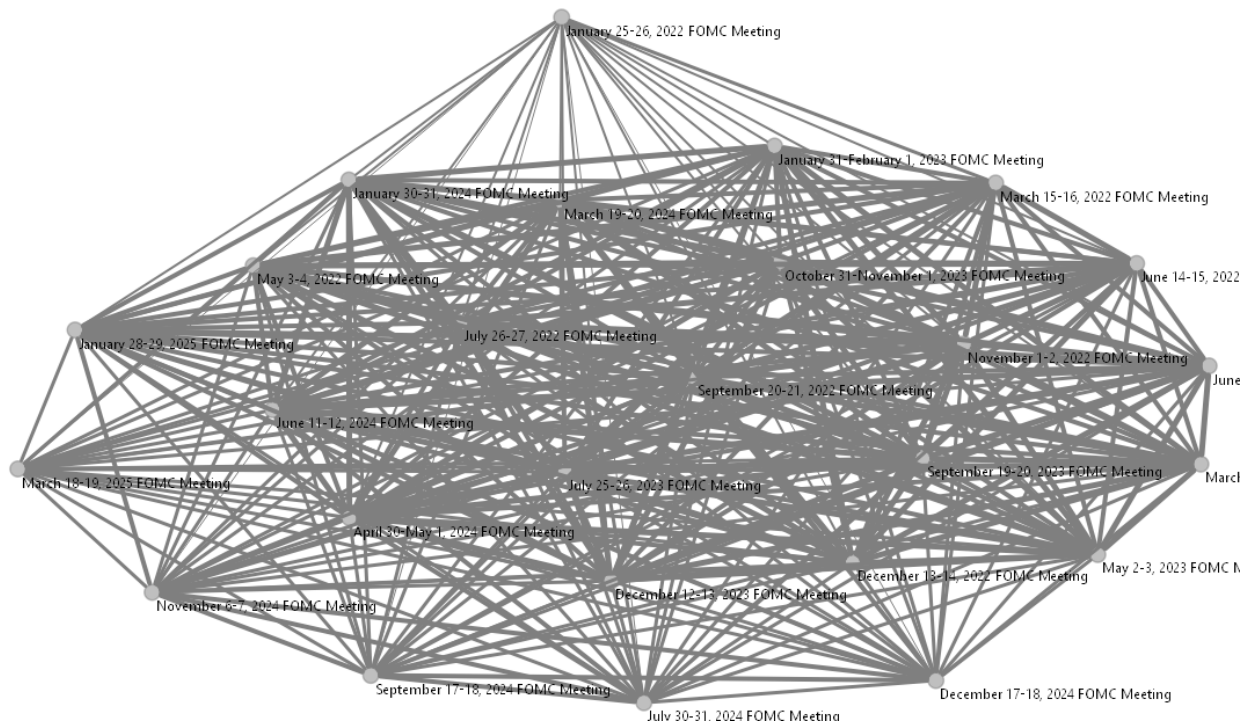
- Approximately 28% of words in each document are unique
- Suggests a controlled vocabulary with repeated terminology used consistently across meetings



Guardian

- The majority of articles contain 90–100% unique words
- This reflects their brevity—typically headlines or summaries—where nearly every word is distinct

Co-occurrence Network – FOMC Minutes (2022–2025)

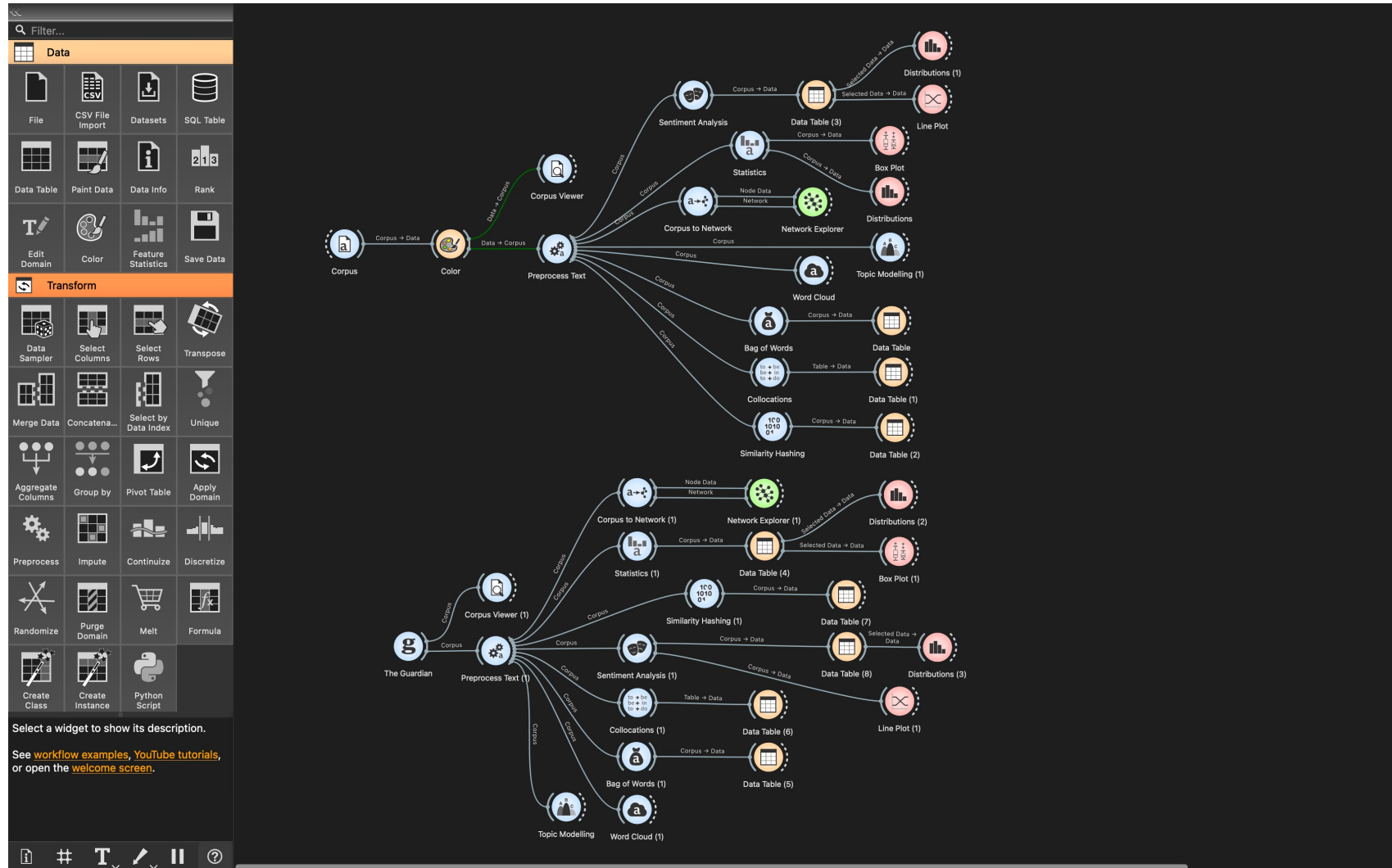


- The network shows dense connectivity, highlighting strong consistency in language, themes, and structure across meetings
- Key terms like “inflation,” “policy,” “market,” and “rates” frequently co-occur, linking meetings closely across time
- The absence of distinct clusters points to a uniform policy communication strategy maintained throughout the period

Final Summary

- FOMC minutes are lengthy, uniform documents characterized by formal language and a structured tone. In contrast, Guardian articles are brief, diverse, and often emotionally charged, reflecting media-driven narratives. Sentiment in FOMC texts tends to be neutral or mildly positive, consistent with cautious policy communication, while Guardian content exhibits strong sentiment swings—frequently leaning negative or overtly positive.
- FOMC discussions revolve around core institutional topics like inflation, interest rates, and policy mechanisms. Meanwhile, Guardian articles emphasize global events, political figures, and public economic issues. The FOMC texts show higher linguistic complexity and frequent term repetition, whereas Guardian content has greater lexical variety due to its concise, unique headlines.
- Network analysis of FOMC texts reveals consistent terminology and messaging over time, highlighting a steady policy voice, in sharp contrast to the dynamic and reactive nature of media discourse.

Orange Workflow



Strategic Recommendations

- Track Sentiment in Real-Time:** Continuously monitor media sentiment alongside FOMC releases to gauge immediate public and market reactions.
- Simplify Policy Communication:** Offer plain-language summaries of FOMC minutes to improve accessibility and foster public understanding.
- Automate Economic Monitoring with NLP:** Deploy NLP workflows (via Orange or Python) to analyze sentiment, topics, and tone during major economic events.
- Identify Emerging Public Concerns:** Use media topic modeling to detect rising themes (like inflation or job security) that could shape economic perception.
- Benchmark Narratives for Alignment:** Regularly compare FOMC messaging with media coverage to identify gaps, misinterpretations, or areas needing clarification.